

```
#install.packages("lmtest")
```

```
library(lmtest)
```

```
library(bestNormalize)
```

```
library(MASS)
```

```
library(tseries)
```

```
library(moments)
```

```
library(dplyr)
```

```
library(ggplot2)
```

```
library(lubridate)
```

```
library(caret)
```

```
library(tseries)
```

```
library(tidyr)
```

```
df1 = Daily_Dataset_dda
```

```
#plot
```

```
hist(df1$`Crude Oil`)
```

```
hist(df1$`Exchange Rate`)
```

```
hist(df1$`Stock price`)
```

```
#outliers
```

```
df_long <- df1 %>%
```

```
  pivot_longer(cols = c(`Crude Oil`, `Stock price`, `Exchange Rate`),
```

```
    names_to = "Variable",
```

```
    values_to = "Value")
```

```
combined_plot <- ggplot(df_long, aes(x = Variable, y = Value, fill = Variable)) +
```

```
  geom_boxplot() +
```

```
  scale_fill_manual(values = c("Crude Oil" = "skyblue", "Stock price" = "lightgreen", "Exchange Rate" =  
"lightpink")) +
```

```

theme_minimal() +
labs(title = "Box Plots of Crude Oil Prices, Stock Prices, and Exchange Rates",
      x = "Variables",
      y = "Value") +
theme(legend.position = "none")
print(combined_plot)

p6 <- ggplot(df, aes(y=`Exchange Rate`)) +
  geom_boxplot(fill="lightpink") +
  theme_minimal() +
  labs(title="Box Plot of Exchange Rates", y="Exchange Rate")
print(p6)

#we can observe from the histogram and the boxplot that there are outliers towards the right end
side.

```

#Skewness Kurtosis

```

skewness_val = apply(df1[,c("Crude Oil", "Stock price", "Exchange Rate")], 2, skewness)
kurtosis_val = apply(df1[,c("Crude Oil", "Stock price", "Exchange Rate")], 2, kurtosis)
skewness_val
kurtosis_val

```

#check correlation

```

corr = cor(df1[,c("Exchange Rate", "Crude Oil", "Stock price")], method = "kendall")
print(corr)

#ask ma'am would multicollinearity play a role in this?

```

#stationary + check ADF test

```

#not using log cz it can affect the outliers or extreme movements to capture them effectively.
df1$log_crude = c(NA, diff(df1$`Crude Oil`))

```

```
plot(df1$Date,df1$log_crude, type = "l")
```

```
df1$log_stock = c(NA,diff(df1$`Stock price`))
```

```
plot(df1$Date,df1$log_stock, type = "l")
```

```
df1$log_exch = c(NA,diff(df1$`Exchange Rate`))
```

```
plot(df1$Date,df1$log_exch, type = "l")
```

```
#adf
```

```
adf.test(log_crude) #pval = 0.01 < 0.05. thus, stationary
```

```
adf.test(na.omit(df1$log_stock)) #pval = 0.01 < 0.05. thus, stationary
```

```
adf.test(na.omit(df1$log_exch)) #pval = 0.01 < 0.05. thus, stationary
```

```
## check with piers prior to maam
```

```
#dw test
```

```
#crude oil autocorrelation
```

```
model <- lm(df1$log_stock ~ df1$log_crude)
```

```
dw_test <- dwtest(model)
```

```
print(dw_test)
```

```
#exch autocorrelation
```

```
model1 = lm(df1$log_stock~df1$log_exch)
```

```
dw_test1 = dwtest(model1)
```

```
print(dw_test1)
```

#GLM.....chosen over lm when applied for marginal dependence on stock_m. since, there is no assumption of linearity under it

```
#converting to uniform using ECDF + plotting
```

```
uni_crude = ecdf(log_crude)
```

```
plot(uni_crude)
```

```
uni_stock = ecdf(df1$log_stock)
```

```
plot(uni_stock)
```

```
uni_exch = ecdf(df1$log_exch)
```

```
plot(uni_exch)
```

```
# Name : Princy Shah
```

```
# DDA project (Survival Analysis part)
```

```
# Part 1 : Kaplan Meier Curve
```

```
# Check the assumptions of Kaplan Meier
```

```
# Edit the dataset to 0 and 1 for relapse
```

```
# find mean median SD with CI & make conclusions
```

```
# The assumptions are as follows,
```

```
# {to be confirmed if the earnings are interpreted correctly}
```

```
# The survival probabilities are same for all the individuals who joined
```

```
# late as well early. ....SATISFIED
```

```
# The occurrence of an event of interest happens at the specified time. ....SATISFIED we have specific dates of relapses.
```

```
# At any time, subjects who are censored have the same survival prospects as
```

```
# those who continue to be followed up. ....SATISFIED
```

```
# Start date of study : 2021-08-04
```

```
# End date of study : 2024-07-25
```

```
# Thus, observations that did not relapse (event) till the end of the study are censored.
```

```
# From domain knowledge we have split alcohol and drug to make significant findings.
```

```
data = final_dataset_relapse_missingvalues_adj
```

```

data$Event = ifelse(data$relapse == "Yes",1,0)

max_value = difftime("2024-08-31","2021-08-04",units = "days")

max_value

# censoring helps is avoiding overestimating the relapse rate.


library(dplyr)

data$`Time till Relapse` = data$`Time till Relapse` %>%
  replace(is.na(.),1123)

#View(data)

# Now, we have both the imp columns the time & event. Thus, moving ahead with KM curve for the
same.


#----- KAPLAN MEIER -----

#Outcome variable

library(survival)

y = Surv(data$`Time till Relapse`,data$Event)

y

curve1 = survfit(y~1,data = data,conf.type = "plain")

str(curve1)

#install.packages("ggsurvfit")

library(ggsurvfit)

??ggsurvfit

survfit2(Surv(data$`Time till Relapse`, data$Event) ~ 1, data = data) %>%

  ggsurvfit() +

  labs(

    x = "Days",

    y = "Overall survival probability for Alcohol addicts",

    color = "Legend",

    title = "Kaplan Meier Curve"

  ) +

  add_confidence_interval() +

```

```
add_risktable() +  
add_censor_mark(shape = 3, size = 2) +  
scale_color_manual(values = "blue", labels = "Alcohol Addiction")  
# observation : there's a sudden drop.
```

```
# yaha legend kyu add nhi ho rha?  
summary(curve1, times = 182.625)  
# 0.907.....6 months  
summary(curve1, times = 365.25)  
# 0.382.....1 yr  
summary(curve1, times = 365.25+182.625)  
# 0.242.....1 yr 6 months
```

CONCLUSION

```
# cumulative survival index showed that in the first six months, about 91% of  
# the subjects did not experience relapse to alcohol,  
# while this index was around 38.2% after a yr.  
# The index drops further to 24.2% in 18 months  
# and was consistent in the next follow up.
```

```
curve1  
# Median = 266 days  
# 266 days is the time corresponding to 0.5 prob.  
# Shorter median -> quicker relapse  
# High short term risk
```

```
summary(curve1)$table[5:6]  
# rmean se(rmean)  
# 472.00311 20.97469  
# on an avg. a person remains alcohol free for 15 months + - 20 days.  
# Mean is the area under the survival curve.
```

Problem : find the date after 266 (median) days of the exit date and understand why the readmission is happening.

#solution :

```
getDaysBetweenDates <- function(start, end, dayName) {  
  result <- c()  
  days <- c(sun=0, mon=1, tue=2, wed=3, thu=4, fri=5, sat=6)  
  day <- days[tolower(substr(dayName, 1, 3))]  
  current <- as.Date(start)  
  end_date <- as.Date(end)  
  shift <- (day - as.integer(format(current, "%w"))) + 7) %% 7  
  current <- current + shift  
  while (current < end_date) {  
    result <- c(result, as.character(current))  
    current <- current + 7  
  }  
  
  return(result)  
}  
  
data$readmission_date = as.Date(data$readmission_date)  
list_sunday = as.Date(getDaysBetweenDates("2021-09-03", "2024-01-21", "Mon"))  
list_sunday[list_sunday %in% data$readmission_date]  
  
# 30% of the people have relapsed on a "Monday" out of the ones that relapsed.  
# Slightly higher than how much ideally it should have been.  
# Before making solid conclusions we need to ask the organisation why this is happening.  
# It could be that people coming to muktangan are from outside thus,  
# taking the weekend to travel.  
  
# All major Indian festival dates (2021-2023) in a single vector  
  
festivals <- as.Date(c(
```

"2021-01-01", "2021-01-14", "2021-01-26", "2021-02-16", "2021-03-11", "2021-03-29",
"2021-04-02", "2021-04-13", "2021-04-14", "2021-04-21", "2021-05-13", "2021-07-21",
"2021-08-15", "2021-08-19", "2021-08-30", "2021-09-10", "2021-10-02", "2021-10-15",
"2021-10-19", "2021-11-04", "2021-11-19", "2021-12-25",

"2022-01-01", "2022-01-14", "2022-01-26", "2022-03-01", "2022-03-18", "2022-04-02",
"2022-04-10", "2022-04-14", "2022-04-15", "2022-05-03", "2022-07-10", "2022-08-11",
"2022-08-15", "2022-08-19", "2022-08-31", "2022-10-02", "2022-10-05", "2022-10-09",
"2022-10-24", "2022-10-26", "2022-11-08", "2022-12-25",

"2023-01-01", "2023-01-15", "2023-01-26", "2023-02-18", "2023-03-08", "2023-03-22",
"2023-04-04", "2023-04-07", "2023-04-14", "2023-04-22", "2023-06-29", "2023-08-15",
"2023-08-30", "2023-09-19", "2023-10-02", "2023-10-24", "2023-10-28", "2023-11-12",
"2023-11-15", "2023-11-27", "2023-12-25"

))

```
festivals[festivals %in% data$readmission_date]
```

```
# Not to be mentioned
```

```
# Can you find if we can potray this in some form of graph?
```

```
df = data.frame()
```

```
#write.csv(data,"C:\\Users\\Hp\\Downloads\\266mediandays.csv", row.names = FALSE)
```

```
# put 2 \\ slashes.
```

```
# limitation : we could have captured more data if the time was longer. 22.2233% had to be censored.
```

```
# -----
```

```
# -----LOG RANK TEST -----
```

```
# Assuming censoring is non - informative (word doc)
```



```

# we are not using the pseudo-code method (stack overflow)

# Merged both the dataset. Since, less obs.


# Student vs Employed

data = log_rank_merged_data

data$Event = ifelse(data$relapse == "Yes",1,0)

max_value = difftime("2024-08-31","2021-08-04",units = "days")

max_value

# censoring helps is avoiding overestimating the relapse rate.


library(dplyr)

library(survival)

library(ggplot2)

library(ggsurvfit)

data$`Time till Relapse` = data$`Time till Relapse` %>%
  replace(is.na(.),1123)

#dataset <- data %>% filter(`Employment Status` == "Student")

filtered_data = data[data$`Employment Status` %in% c("Employed","Student"), c("Time till
Relapse","Event","Employment Status")]

curve2 = survfit(Surv(filtered_data$`Time till Relapse`,filtered_data$Event) ~
filtered_data$`Employment Status`,data = filtered_data,conf.type = "plain")

curve2 %>%

  ggsurvfit() +

  labs(

    x = "Days",

    y = "Overall survival probability of addicts",

    color = "Legend",

    title = "Log Rank test for Employed and Student"

  ) +

  add_confidence_interval() +

  add_censor_mark(shape = 3, size = 2) +

  scale_color_manual(values = c("red", "blue"), labels = c("employed", "Student"))

```

```

# The drop suggests students have a quicker relapse compared to Employed people

curve2

survdifff(Surv(filtered_data$`Time till Relapse`, filtered_data$Event) ~ filtered_data$`Employment
Status`, data = filtered_data)

# Employed have a gradual relapse rate
# Student have a quicker relapse rate
# The p-value suggests there is a significant difference between both the categories.

summary(curve2)$table

# mean relapse rate of employed = 14 months +- 27 days
# mean relapse rate of student = 7 months +- 18 days
# # Majority of students are relapsing in autumn & winter season.
# Majority of Employed are relapsing in Winters. (Pune : 11'C - 28'C)
# Could be due to cold.
# -----

# -----COX REGRESSION -----

data = final_dataset_relapse_missingvalues_adj
data$Event = ifelse(data$relapse == "Yes",1,0)
data$`Time till Relapse` = data$`Time till Relapse` %>%
  replace(is.na(.),1123)

# Do I need to change my categorical variables and mae 6 separate columns based on it?
# As of now I have not done the same.

data$isolation = ifelse(data$`Marital Status` == "M",0,1)
data$employ = ifelse(data$`Employment Status` == "Employed",1,
  ifelse(data$`Employment Status` == "N.E.",2,
    ifelse(data$`Employment Status` == "Part Time",3,
      ifelse(data$`Employment Status` == "Pensioner",4,
        ifelse(data$`Employment Status` == "S.E.",5,
          ifelse(data$`Employment Status` == "Employed",6,
            ifelse(data$`Employment Status` == "U.E.",7,8)))))))))

```

```

#View(data) # 8 = student
data$qual <- ifelse(data$Qualification == "Graduate", 1,
  ifelse(data$Qualification == "H.S.C.", 2,
    ifelse(data$Qualification == "Masters", 3,
      ifelse(data$Qualification == "Middle School", 4, 5))))

# Univariate cox does not take into ac the interactionterm. Thus, terrible choice.
# Stepwise regression
# MCP
# Multicollinearity

data$employ = as.factor(data$employ)
data$qual= as.factor(data$qual)
data$isolation = as.factor(data$isolation)
#data$annual =
# data$Age = scale(data$Age)
# stepwise regression
res.cox4 <- coxph(Surv(data$`Time till Relapse`, data$Event) ~
  data$Age+data$isolation+data$qual+data$employ+data$`Annual Income`, data = data)
summary(res.cox4)
final <- step(res.cox4, direction="both")
summary(final)

# Employ status was removed during the process
# Isolation increases the chances of relapse by 63.34%
# likelihood test , Wald's test suggest model is statistically significant

#-----ASSUMPTIONS OF COX-----

# To test for the proportional-hazards (PH) assumption
test.ph = cox.zph(res.cox4)
test.ph

#The global test is not statistically significant

```

```

# thus we can assume proportional hazards.

library(survminer)

ggcoxzph(test.ph)

#all the lines look pretty straight to me. Thus, no pattern is formed with time.


# Testing influential observations

ggcoxdiagnostics(res.cox4, type = "dfbeta",

                  linear.predictions = FALSE, ggtheme = theme_bw())

# dfBeta >0.2 for any pt indicates influential obs.

## almost all these variables are categorical. Do they still need treatment?

# it's just the composition of your sample, not an outlier.


# Non - linearity

# Assumption is satisfied as it lies in a band.

cox_model <- coxph(Surv(`Time till Relapse`, Event) ~ Age + log(Age) + sqrt(Age) + `Annual Income` +
log(`Annual Income`) + sqrt(`Annual Income`), data = data)

par(mfrow = c(2,2))

plots <- ggcoxfunctional(fit = cox_model, data = data)

plots <- lapply(plots, function(p) p + coord_cartesian(ylim = c(-1.715644, 1.715644)))

plots

summary(cox_model)

sd(cox_model$residuals)

# Upper limit

upper = 2*0.8671383

lower = -2*0.8671383

# 1.734277


# AIC value

# AIC = -2*(partial log-likelihood)+ 2*k

edf = length(final$coefficients)

-2*logLik(final)+2*edf

```

```

#-----

#-----MULTICOLLINEARITY, VIF, R^2-----

# Multicollinearity
# "Rsquare" value because it is not correct under right censoring
# Nagelkerke's R2 approximation
n = 322
x = 18.81
Rsqr = 1 - exp((-x)/n)
Rsqr

# since, confounders are missing it captures small fraction of variability.
# Thus, need for better predictors.


# Function to calculate VIF for Cox model
compute_vif_cox <- function(res.cox4, data) {
  vif_values <- c()
  predictors <- names(res.cox4$coefficients) # Extract predictor names

  for (var in predictors) {
    # Fit a regression model predicting 'var' using all other predictors
    formula_str <- paste(var, "~", paste(setdiff(predictors, var), collapse = " + "))
    model <- lm(as.formula(formula_str), data = data)

    # Compute R^2 for this model
    r2 <- summary(model)$r.squared

    # Compute VIF
    vif_values[var] <- 1 / (1 - Rsqr)
  }
}

```

```
    return(vif_values)
}
```

```
# Example usage
```

```
res.cox5 <- coxph(Surv(data$`Time till Relapse`, Event) ~ Age + data$`Annual Income`, data = data)
vif_results <- compute_vif_cox(res.cox5,data)
print(vif_results)
```

```
library(car)
```

```
model1 = lm()
```

```
# stigma people are not there to talk to.
```

```
# relatives dont want to talk to rehab people.
```

```
res.cox5 <- coxph(Surv(data$`Time till Relapse`, Event) ~ Age + data$`Annual Income`, data = data)
```

```
model1 = lm(`Time till Relapse`~ Age + `Annual Income` + isolation + qual + employ, data = data)
```

```
library(car)
```

```
values = vif(model1)
```

```
values
```

```
res.cox4 <- coxph(Surv(data$`Time till Relapse`, data$Event) ~
data$Age+data$isolation+data$qual+data$employ+data$`Annual Income`, data = data)
```

```
#-----CENSORED KA CLUSTER -----
```